

Design of Counters

This example is taken from M. M. Mano, *Digital Design*, Prentice Hall, 1984, p.243.

Example 1.6 Design a counter specified by the state diagram in [Example 1.5](#) using T flip-flops. The state diagram is shown here again in Figure 22.

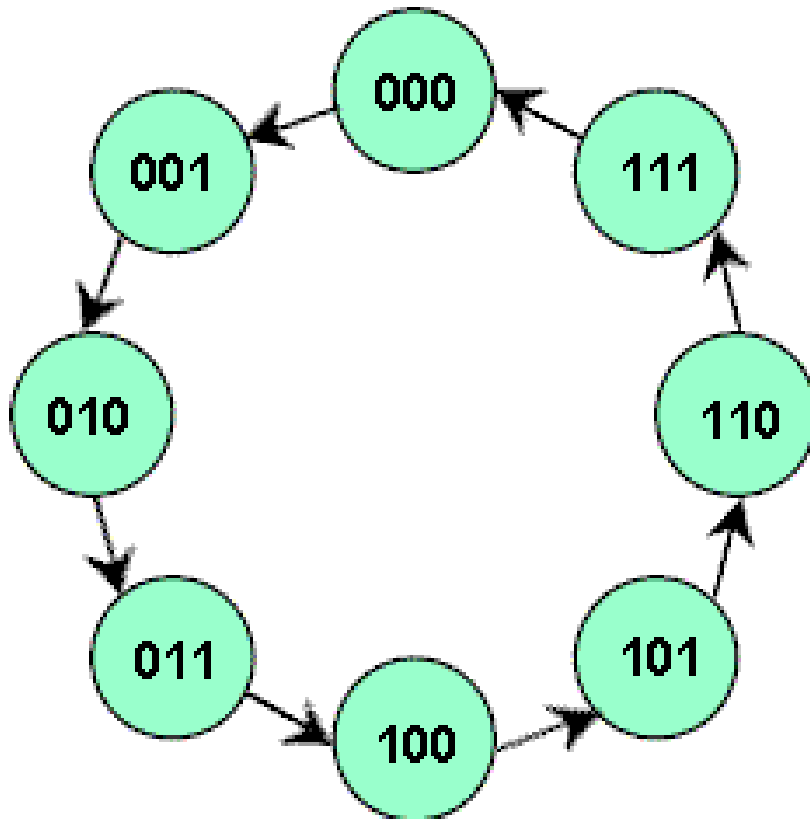


Figure 22.
State
diagram
of a 3-bit
binary
counter.

The state table will be the same as in Example 1.5.

Now derive the excitation table from the state table, which is shown in Table 17.

Table 17. Excitation table.

Output State Transitions			Flip-flop inputs		
Present State $Q_2 \ Q_1 \ Q_0$			Next State $Q_2 \ Q_1 \ Q_0$		
			$T_2 \ T_1 \ T_0$		
0	0	0	0	0	1
0	0	1	0	1	1
0	1	0	0	0	1
0	1	1	1	1	1
1	0	0	0	0	1
1	0	1	0	1	1
1	1	0	0	0	1
1	1	1	1	1	1

Next step is to transfer the flip-flop input functions to Karnaugh maps to derive a simplified Boolean expressions, which is shown in Figure 23.

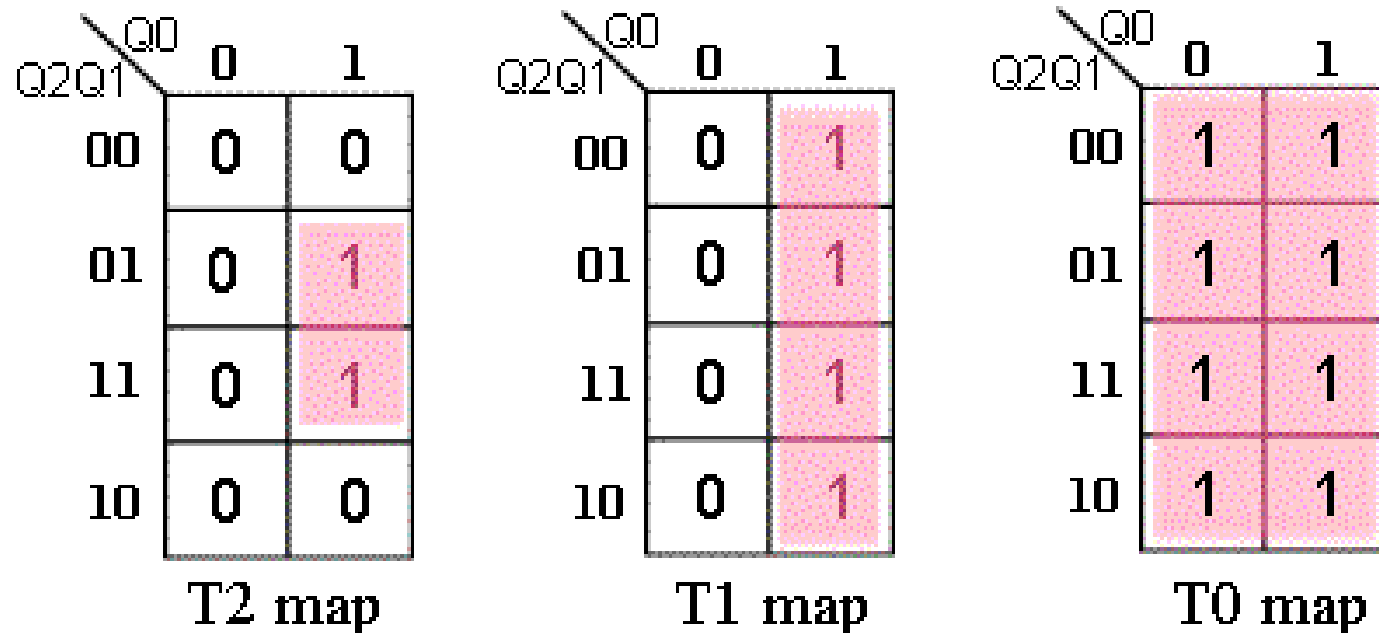


Figure 23.
Karnaugh
maps

The following expressions are obtained:

$$T_0 = 1; \quad T_1 = Q_0; \quad T_2 = Q_1 * Q_0$$

Finally, draw the logic diagram of the circuit from the expressions obtained. The complete logic diagram of the counter is shown in Figure 24.

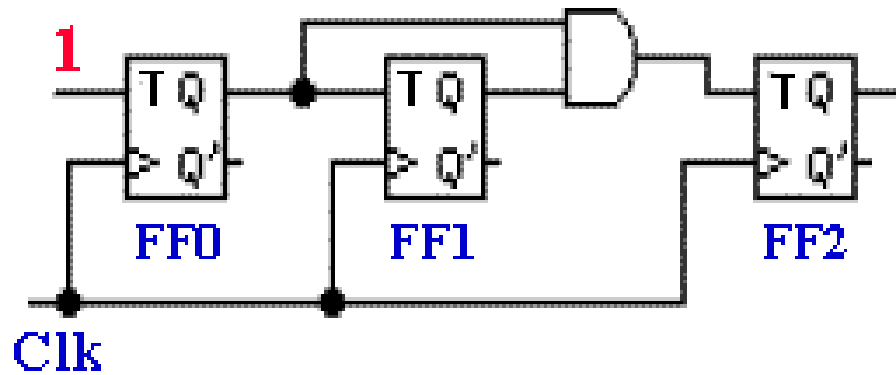
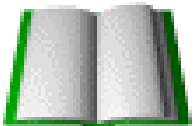


Figure 24. Logic diagram of 3-bit binary counter.

To see the timing and state transitions of the counter, click on this image. 

Now that you have reached the end of the tutorial, you should be able to understand the basic concept of sequential circuits. You should be able to analyze and design a basic sequential circuit. Now you can practice some of the exercises using the analysis and design procedures shown in the examples.



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